

The very first experiment:

QM2D (348 train, 872 test, overall: 4357 planar molecules). Only valence. Normalized y. Training iterations = 500.					
Vanilla GP			Scatter GP (J=3, L=8)		
	rbf	rbf, ls=sqrt(dim)	rbf-ard, ls=sqrt(dim)	rbf	rbf-ard
RMSE	1.00	0.73	0.73	0.60	0.73
MAE	0.77	0.57	0.57	0.47	0.53
NLL	1.39	1.05	1.05	0.77	1.65
Time (s)	9	9	23	53	1879

Here we consider a small-ish train set and one channel (valence), and evaluate Scatter with different J, rotation invariance and number of training iterations:

QM2D (697 train, 436 test, overall: 4357 planar molecules). Only valence. Normalized y. Training iterations = 500.													
Vanilla GP				Scatter GP (J=8, L=8)		Scatter GP (J=7, L=8)		Scatter GP (J=6, L=8)		Scatter GP (J=5, L=8)		Scatter GP (J=4, L=8)	
	rbf	rbf, ls=sqrt(dim)	rbf-ard, ls=sqrt(dim)	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard
RMSE	0.87	0.70	0.70	0.27	0.11	0.30	0.15	0.31	0.18	0.42	0.32	0.55	0.55
RMSE (unnormalized)	154.5	124.3	124.3	48.0	19.5	53.0	25.9	54.6	31.6	75.0	56.4	98.4	97.0
MAE	0.66	0.54	0.54	0.21	0.08	0.22	0.11	0.23	0.13	0.33	0.23	0.42	0.36
MAE (unnormalized)	117.2	95.9	95.9	37.3	14.2	39.5	18.8	40.8	23.4	59.3	40.5	75.2	64.3
NLL	1.20	1.04	1.04	-0.1	-0.75	-0.02	0.02	0.01	0.84	0.44	2.55	0.69	1.72
Time (s)	19	19	54	2	3	3	6	6	15	14	37	31	92
# features	262144			7428 = 1857 * 2 * 2		22416 = 1401 * 4 * 4		64576 = 1009 * 8 * 8		174336 = 681 * 16 * 16		427008 = 417 * 32 * 32	

QM2D (697 train, 436 test, overall: 4357 planar molecules). Only valence. Normalized y. Training iterations = 10000.													
Vanilla GP				Scatter GP (J=8, L=8)		Scatter GP (J=7, L=8)		Scatter GP (J=6, L=8)		Scatter GP (J=5, L=8)		Scatter GP (J=4, L=8)	
	rbf	rbf, ls=sqrt(dim)	rbf-ard, ls=sqrt(dim)	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard
RMSE	0.57	0.58	0.7	0.22	0.12	0.24	0.15	0.23	0.64	0.32		0.55	0.72
RMSE (unnormalized)	100.8	102.8	125.0	38.2	20.6	41.9	26.1	40.4	113.5	56.8		98.4	128.5
MAE	0.44	0.45	0.54	0.16	0.08	0.17	0.11	0.17	0.48	0.25		0.42	0.39
MAE (unnormalized)	78.5	80.7	95.3	28.8	14.6	30.9	19.1	29.5	84.5	44.1		75.2	70.0
NLL	0.76	0.85	2.2	-0.44	1.19	-0.36	2.27	-0.4	2.06	0.1		0.69	11.48
Time (s)	362	369	1050	43	13	64	19	118	296	271		614	563
# features	262144			7428 = 1857 * 2 * 2		22416 = 1401 * 4 * 4		64576 = 1009 * 8 * 8		174336 = 681 * 16 * 16		427008 = 417 * 32 * 32	

QM2D (697 train, 436 test, overall: 4357 planar molecules). Only valence. Normalized y. Training iterations = 500. Rotation-invariant features.											
Vanilla GP			Scatter GP (J=8, L=8)		Scatter GP (J=7, L=8)		Scatter GP (J=6, L=8)		Scatter GP (J=5, L=8)		
	rbf	rbf, ls=sqrt(dim)	rbf-ard, ls=sqrt(dim)	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard
RMSE	See above			0.29	0.2	0.28	0.17	0.3	0.24	0.39	0.38
RMSE (unnormalized)				51.0	34.9	50.5	30.1	52.6	43.2	69.5	67.3
MAE				0.22	0.1	0.21	0.13	0.22	0.17	0.3	0.27
MAE (unnormalized)				39.5	18.2	38.1	22.5	39.6	30.5	53.4	47.4
NLL				0.05	-0.78	-0.02	-0.54	-0.03	-0.3	0.3	0.31
Time (s)				2	2	2	2	2	3	3	6
# features				932 = 233 * 2 * 2		2816 = 176 * 4 * 4		8128 = 127 * 8 * 8		22016 = 86 * 16 * 16	

QM2D (697 train, 436 test, overall: 4357 planar molecules). Only valence. Normalized y. Training iterations = 10000. Rotation-invariant features.											
Vanilla GP			Scatter GP (J=8, L=8)		Scatter GP (J=7, L=8)		Scatter GP (J=6, L=8)		Scatter GP (J=5, L=8)		
	rbf	rbf, ls=sqrt(dim)	rbf-ard, ls=sqrt(dim)	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard
RMSE	See above			0.22	0.11	0.22	0.13	0.22		0.3	
RMSE (unnormalized)				38.3	19.0	38.8	22.3	38.3		52.5	
MAE				0.17	0.08	0.17	0.09	0.16		0.22	
MAE (unnormalized)				29.4	14.3	29.7	16.7	27.7		39.4	
NLL				-0.32	-0.86	-0.34	-0.14	-0.37		-0.02	
Time (s)				38	39	38	45	44		62	
# features				932 = 233 * 2 * 2		2816 = 176 * 4 * 4		8128 = 127 * 8 * 8		22016 = 86 * 16 * 16	

Here we consider increasing training data size and compare different J and rotation-invariant vs vanilla scattering features:

QM2D With Dropped Duplicates (840 train (0.2), 420 test, overall: 4201 planar molecules). Only valence. Normalized y. Training iterations = 500. Initial lengthscale = mean(dists). Initial noise = 0.1. 5 repeats. PCA-0.99.										
Vanilla GP		Scatter GP (J=8, L=8)		Scatter GP (J=8, L=8) ROT-INV		Scatter GP (J=7, L=8)		Scatter GP (J=7, L=8) ROT-INV		
	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard
RMSE	0.54±0.03	0.57±0.02	0.19±0.02	0.23±0.0	0.2±0.02	0.13±0.01	0.21±0.02	0.28±0.03	0.21±0.01	0.21±0.01
RMSE (unnormalized)	97.76±5.14	104.55±5.07	35.46±3.69	42.93±0.82	37.21±4.07	23.23±1.24	38.07±2.45	50.5±4.73	37.72±2.61	38.71±1.69
MAE	0.42±0.02	0.44±0.01	0.14±0.02	0.18±0.0	0.15±0.01	0.1±0.0	0.15±0.01	0.21±0.02	0.15±0.01	0.16±0.01
MAE (unnormalized)	75.55±3.56	80.97±2.69	26.01±2.53	33.33±1.17	27.55±1.96	18.05±0.86	26.93±0.38	38.98±3.48	27.04±1.64	29.43±1.68
NLL	0.75±0.03	0.85±0.06	-0.49±0.06	13.09±1.51	-0.32±0.05	0.2±0.16	-0.45±0.04	16.34±1.24	-0.37±0.05	6.22±1.15
Time (s)	14.5±0.08	27.64±0.08	14.53±0.03	27.87±0.05	9.4±0.02	12.65±0.02	14.49±0.02	27.68±0.03	12.33±0.03	21.57±0.1
QM2D With Dropped Duplicates (1260 train (0.3), 420 test, overall: 4201 planar molecules). Only valence. Normalized y. Training iterations = 500. Initial lengthscale = mean(dists). Initial noise = 0.1. 5 repeats. PCA-0.99.										
Vanilla GP		Scatter GP (J=8, L=8)		Scatter GP (J=8, L=8) ROT-INV		Scatter GP (J=7, L=8)		Scatter GP (J=7, L=8) ROT-INV		
	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard
RMSE	0.5±0.02	0.51±0.02	0.15±0.01	0.18±0.01	0.17±0.01	0.1±0.01	0.15±0.01	0.21±0.01	0.16±0.01	0.15±0.01
RMSE (unnormalized)	92.24±3.74	94.3±3.17	27.41±1.4	32.69±1.58	30.55±1.51	17.53±1.45	27.97±1.59	38.51±1.65	28.57±1.88	26.44±1.13
MAE	0.39±0.02	0.4±0.02	0.11±0.0	0.14±0.01	0.13±0.01	0.07±0.0	0.11±0.01	0.16±0.01	0.12±0.01	0.11±0.01
MAE (unnormalized)	71.2±2.45	72.73±2.59	19.75±0.69	25.08±1.1	22.84±1.08	13.15±0.85	20.51±0.73	29.36±1.22	21.32±1.32	19.98±1.0
NLL	0.7±0.04	0.73±0.05	-0.71±0.04	6.44±1.74	-0.05±0.03	-0.68±0.2	-0.69±0.04	9.61±1.25	-0.57±0.06	2.15±0.29
Time (s)	28.08±0.03	54.09±0.09	28.1±0.03	53.92±0.19	17.25±0.07	23.1±0.25	28.07±0.05	54.29±0.12	23.64±0.05	40.5±0.2
QM2D With Dropped Duplicates (1680 train (0.4), 420 test, overall: 4201 planar molecules). Only valence. Normalized y. Training iterations = 500. Initial lengthscale = mean(dists). Initial noise = 0.1. 5 repeats. PCA-0.99.										
Vanilla GP		Scatter GP (J=8, L=8)		Scatter GP (J=8, L=8) ROT-INV		Scatter GP (J=7, L=8)		Scatter GP (J=7, L=8) ROT-INV		
	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard
RMSE	0.48±0.01	0.48±0.02	0.13±0.1	0.15±0.01	0.14±0.0	0.08±0.01	0.13±0.0	0.17±0.01	0.15±0.0	0.12±0.01
RMSE (unnormalized)	86.46±1.58	86.34±3.33	23.87±1.73	27.54±1.01	26.25±0.79	14.65±1.05	23.71±0.61	30.35±0.84	26.85±0.77	21.66±1.93
MAE	0.37±0.01	0.36±0.02	0.09±0.01	0.11±0.01	0.11±0.0	0.06±0.0	0.09±0.0	0.13±0.01	0.11±0.0	0.09±0.01
MAE (unnormalized)	66.37±0.94	65.8±2.0	16.92±1.28	20.51±0.79	19.78±0.05	10.93±0.58	16.63±0.46	23.09±0.67	19.19±0.19	15.97±0.4
NLL	0.65±0.02	0.62±0.06	-0.88±0.06	4.05±0.84	-0.64±0.02	-1.05±0.13	-0.89±0.05	5.8±0.49	-0.7±0.03	0.57±0.4
Time (s)	53.34±0.08	97.58±0.17	53.33±0.09	96.63±0.1	34.54±0.07	43.57±0.12	53.32±0.08	96.94±0.45	45.43±0.07	74.63±0.05
QM2D With Dropped Duplicates (2100 train (0.5), 420 test, overall: 4201 planar molecules). Only valence. Normalized y. Training iterations = 500. Initial lengthscale = mean(dists). Initial noise = 0.1. 5 repeats. PCA-0.99.										
Vanilla GP		Scatter GP (J=8, L=8)		Scatter GP (J=8, L=8) ROT-INV		Scatter GP (J=7, L=8)		Scatter GP (J=7, L=8) ROT-INV		
	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard
RMSE	0.47±0.03	0.47±0.02	0.11±0.01	0.13±0.0	0.13±0.01	0.07±0.0	0.11±0.01	0.14±0.01	0.12±0.01	0.09±0.01
RMSE (unnormalized)	85.83±3.46	87.1±3.46	19.75±1.15	23.19±0.9	24.04±1.5	12.99±0.18	20.47±1.09	26.51±1.65	22.25±1.06	16.65±1.24
MAE	0.36±0.02	0.35±0.01	0.08±0.01	0.1±0.0	0.1±0.01	0.05±0.0	0.08±0.0	0.11±0.01	0.09±0.0	0.07±0.0
MAE (unnormalized)	65.19±2.03	65.45±2.86	14.26±0.85	17.52±0.89	17.86±0.95	9.87±0.27	14.44±0.66	20.07±1.36	16.26±0.41	12.49±0.84
NLL	0.62±0.06	0.62±0.05	-1.06±0.05	2.76±0.72	-0.75±0.04	-1.23±0.07	-1.03±0.04	3.77±0.64	-0.83±0.02	-0.44±0.26
Time (s)	86.45±0.04	154.3±0.14	86.48±0.05	154.02±0.24	58.39±0.06	71.83±0.11	86.46±0.05	154.4±0.18	74.62±0.04	118.37±0.26
QM2D With Dropped Duplicates (2520 train (0.6), 420 test, overall: 4201 planar molecules). Only valence. Normalized y. Training iterations = 500. Initial lengthscale = mean(dists). Initial noise = 0.1. 5 repeats. PCA-0.99.										
Vanilla GP		Scatter GP (J=8, L=8)		Scatter GP (J=8, L=8) ROT-INV		Scatter GP (J=7, L=8)		Scatter GP (J=7, L=8) ROT-INV		
	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard
RMSE	0.43±0.03	0.44±0.03	0.1±0.0	0.11±0.11	0.13±0.01	0.06±0.0	0.1±0.01	0.12±0.0	0.11±0.01	0.08±0.0
RMSE (unnormalized)	78.74±5.46	79.49±5.36	17.62±0.8	20.22±1.47	23.62±1.65	11.79±0.66	18.44±1.25	22.04±0.67	20.73±1.1	14.62±0.62
MAE	0.32±0.02	0.32±0.01	0.07±0.0	0.08±0.01	0.09±0.01	0.05±0.0	0.07±0.0	0.09±0.0	0.08±0.0	0.06±0.0
MAE (unnormalized)	58.58±3.63	59.08±2.64	12.07±0.4	15.04±1.12	17.03±0.93	8.77±0.33	12.62±0.56	16.7±0.67	14.56±0.2	10.68±0.46
NLL	0.51±0.06	0.51±0.06	-1.19±0.05	1.58±0.53	-0.81±0.05	-1.39±0.06	-1.18±0.03	2.18±0.37	-0.95±0.02	-0.82±0.13
Time (s)	129.6±0.04	224.37±0.38	129.61±0.02	223.96±0.3	89.51±0.02	107.28±0.2	129.63±0.01	224.17±0.37	112.56±0.03	174.22±0.11
QM2D With Dropped Duplicates (2940 train (0.7), 420 test, overall: 4201 planar molecules). Only valence. Normalized y. Training iterations = 500. Initial lengthscale = mean(dists). Initial noise = 0.1. 5 repeats. PCA-0.99.										
Vanilla GP		Scatter GP (J=8, L=8)		Scatter GP (J=8, L=8) ROT-INV		Scatter GP (J=7, L=8)		Scatter GP (J=7, L=8) ROT-INV		
	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard
RMSE	0.42±0.02	0.43±0.01	0.09±0.01	0.1±0.01	0.12±0.0	0.06±0.0	0.09±0.01	0.11±0.0	0.1±0.01	0.07±0.0
RMSE (unnormalized)	75.44±2.73	78.98±2.28	17.23±2.5	17.5±1.44	22.29±0.85	11.51±0.74	16.82±1.45	20.21±0.47	18.82±1.57	13.08±0.73
MAE	0.31±0.01	0.31±0.01	0.06±0.0	0.07±0.0	0.09±0.0	0.05±0.0	0.06±0.0	0.08±0.0	0.07±0.0	0.05±0.0
MAE (unnormalized)	55.93±1.64	57.91±1.45	11.07±0.72	12.94±0.85	15.78±0.19	8.44±0.5	11.4±0.46	15.08±0.32	13.38±0.73	9.4±0.29
NLL	0.48±0.04	0.5±0.04	-1.31±0.03	0.76±0.18	-0.86±0.02	-1.42±0.09	-1.31±0.04	1.47±0.18	-1.03±0.05	-1.15±0.1
Time (s)	190.34±0.06	314.4±1.02	190.28±0.03	312.99±0.32	136.54±0.06	159.82±0.78	190.13±0.04	314.25±1.04	167.6±0.09	248.94±1.09

QM2D With Dropped Duplicates (3360 train (0.8), 420 test, overall: 4201 planar molecules). Only valence. Normalized y. Training iterations = 500. Initial lengthscale = mean(dists). Initial noise = 0.1. 5 repeats. PCA-0.99.																	
	Vanilla GP		Scatter GP (J=8, L=8)		Scatter GP (J=8, L=8) ROT-INV		Scatter GP (J=7, L=8)		Scatter GP (J=7, L=8) ROT-INV								
	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard	rbf	rbf-ard							
RMSE	0.39±0.02	0.41±0.01	0.08±0.0	0.08±0.0	0.11±0.01	0.06±0.0	0.09±0.01	0.1±0.0	0.1±0.0	0.07±0.0							
RMSE (unnormalized)	72.19±3.92	74.29±2.46	14.23±0.83	15.46±0.68	20.4±1.15	10.47±0.5	15.72±2.08	17.96±0.8	17.68±0.57	12.12±0.49							
MAE	0.29±0.01	0.3±0.01	0.06±0.0	0.06±0.0	0.08±0.0	0.04±0.0	0.06±0.0	0.07±0.0	0.07±0.0	0.05±0.0							
MAE (unnormalized)	52.37±2.25	54.54±1.73	10.0±0.5	11.3±0.45	14.43±0.62	7.74±0.31	10.29±0.66	13.23±0.58	12.34±0.61	8.81±0.24							
NLL	0.39±0.05	0.41±0.05	-1.4±0.05	0.02±0.14	-0.92±0.05	-1.52±0.06	-1.42±0.02	0.61±0.18	-1.13±0.03	-1.28±0.05							
Time (s)	264.11±0.05	420.74±0.05	264.14±0.04	431.58±2.26	194.04±0.02	235.55±0.04	264.12±0.04	423.31±2.77	234.56±0.07	352.27±1.57							